

Rossini Martyr

Data Science | Analytics | Engineering

647-938-9055 | rossinimartyr@gmail.com | linkedin.com/in/rossinimartyr/ | <https://rossini-martyr.github.io/>

MOTIVATION

I am a professional engineer, driven to **solve business problems** using Data Science, Analytics, and Machine Learning. I bring a systematic and collaborative approach to **add tangible value** to the team, business, and clients. I am a **life-long learner**, constantly seeking new ways to improve and innovate.

SKILLS & TOOLS

Programming: SQL, Python (Pandas, Numpy, Matplotlib, Scikit-Learn, Langchain)

Tools: Excel, Tableau, Github, Streamlit, AWS SageMaker, AutoML

Math: Statistics (Hypothesis Testing, AB Testing, Causal Inference), Linear Algebra

Machine Learning: Linear Regression, Logistic Regression, Decision Trees, Random Forest, KNN, k-means, PCA, Gradient Boosted Trees, Retrieval Augmented Generation

PROJECTS

Credit Card Transaction Fraud Detection ([Github](#))

- To minimize financial losses from credit card fraud, I built an **XGBoost classifier** on a highly imbalanced dataset of credit card transactions (578:1). Achieving a **Precision-Recall AUC** of 0.865, this model could be deployed to automatically block fraudulent charges while minimizing customer friction. I optimized parameters using **3-fold cross-validation**, demonstrating that supervised algorithms drastically outperform unsupervised anomaly detection for this application.

Insurance Flood Damage Prediction ([Github](#))

- To quantify financial exposure due to flooding, I built a **Random Forest regression model** to predict flood damage using insurance claim data. I optimized model parameters using cross validation and used SHAP to determine relative feature importance. I deployed the model using Streamlit to allow the client to evaluate the expected damage to **+/- 15%** for different property characteristics.

Quantifying Sales Uplift using Causal Impact Analysis ([Github](#))

- To quantify the financial uplift of a new membership tier, I conducted a **Causal Impact Analysis** to measure if waiving per-order delivery fees increased overall customer spending for a grocery retailer. I compared members against a control group of non-members using three months of pre- and post-campaign transaction data. The analysis revealed a statistically significant (95%) sales uplift of 41.1% among club members, suggesting that eliminating delivery friction drove higher overall revenue.

Assessing Campaign Performance Using A/B Testing ([Github](#))

- To optimize campaign ROI, I conducted an A/B test to determine if expensive promotional mailers drove significantly higher signup rates than cheaper alternatives. Utilizing a Chi-Square Test for Independence, I showed that while the expensive mailer yielded a higher raw signup rate (37.8% vs. 32.8%), the difference was not statistically significant (p-value: 0.16). Therefore, campaign funds could be redirected since the more expensive mailers did not conclusively guarantee higher signup rates.

WORK EXPERIENCE

Senior Engineer, Fast + Epp (Toronto, Canada)

February 2024 – Present

- Project lead for the technical delivery of the \$120M New Brunswick Museum project, optimizing structural solutions to save over \$200,000 in material and resource efficiency
- Managed cross-functional teams on \$66M infrastructure projects, meeting stakeholder requirements while ensuring on-time and on-budget delivery of key deliverables
- Mentored junior and intermediate engineers, providing career guidance and subject matter review

Senior Engineer, Arup (Toronto, Los Angeles)

April 2016 – February 2024

- Developed an SQL and Python-based workflow to analyze datasets with millions of rows, leveraging Amazon cloud compute power to accelerate and iterate design solutions on hospital in Turkey
- Automated the creation of analytical models of underground stations in Toronto using Python-based API's, improving efficiency and accuracy by approximately 30%
- Supervised multiple junior engineers, conducting annual reviews, reviewing peer feedback and providing recommendations on career direction and promotion readiness

EDUCATION

MASc (Structural Engineering) (GPA: 3.94/4.00)

2013 - 2016

University of Toronto

- Published M.A.Sc research in prestigious structural engineering journal

BASc (Civil Engineering) (GPA: 3.88/4.00)

2008 - 2013

University of Toronto

- Graduated #1 in Civil Engineering

COURSES, CERTIFICATES AND QUALIFICATIONS

Data Science & AI Professional Certification

2026 - Ongoing

Data Science Infinity

- **Actionable Learnings:** Advanced data visualizations using Tableau. Designing, executing, and evaluating rigorous A/B tests and causal impact analyses to measure business impact and drive data-informed product decisions. Packaging machine learning models into production-ready APIs and deploying them to AWS cloud environments using SageMaker and AutoML. Building production style AI assistants using Retrieval Augmented Generation (RAG).

Data Science Professional Certification

2025 - 2026

IBM

- **Actionable Learnings:** Querying databases and manipulating data structures using SQL. Automated web scraping using BeautifulSoup and web APIs. Utilizing Pandas and NumPy for data wrangling, and Matplotlib, Folium, and Plotly to build interactive geospatial and statistical visualizations. Implementing regression, classification, and clustering algorithms using Scikit-learn. Navigating GitHub for version control. Leveraging GenAI tools to accelerate code generation, debugging, and data exploration.

Professional Engineer in Ontario

2018 - Current

PEO

- **Actionable Learnings:** As a licensed professional engineer, my foremost responsibility is to apply rigorous engineering methodology to ensure public safety in buildings and infrastructure projects.